

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 11 — Environmental Chemistry: Air

ANSWER KEY (Version 1)

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. The approximate percentage of nitrogen in clean, dry air is: ✓ **C — 78%** **1 mark**

Clean, dry air is approximately **78% nitrogen**, 21% oxygen, and the remainder is noble gases and carbon dioxide. Option A (21%) is the percentage of oxygen, a common confusion. Argon accounts for approximately 0.93%.

Q2. Sulphur dioxide is produced mainly from: ✓ **B — burning of fossil fuels containing sulphur**

1 mark

Sulphur is found naturally in fossil fuels. Burning these fuels in power plants, vehicles, and industrial units releases significant amounts of SO₂. Decomposition of vegetation produces methane, not SO₂. Photosynthesis consumes CO₂ rather than producing pollutants.

Q3. Carbon monoxide is dangerous because it: ✓ **B — combines with haemoglobin to form carboxyhaemoglobin** **1 mark**

CO is colourless and **odourless**, so it cannot be detected by smell. It combines with **haemoglobin** to form carboxyhaemoglobin, which cannot carry oxygen, leading to suffocation and death. It does not cause acid rain — that is caused by SO₂ and NO_x.

Q4. NO₂ is described as a: ✓ **B — reddish-brown gas with pungent odour** **1 mark**

The textbook states: **NO₂ is a reddish-brown gas with pungent odour** soluble in water. NO (option A description) is colourless and odourless. SO₂ is colourless with an irritating odour — a common distractor confusion.

Q5. Greenhouse gases act like a blanket by: ✓ **B — absorbing and re-radiating infrared radiation**

1 mark

When the Earth's surface emits energy as **infrared radiation**, greenhouse gases absorb this energy and re-radiate some of it back towards Earth, trapping heat in the lower atmosphere. They do not reflect all sunlight back — visible light passes through. Filtering UV is the role of stratospheric ozone, a different topic.

Q6. Acid rain is defined as rain with a pH: ✓ **C — less than 5.6** **1 mark**

Acid rain is defined as rain having pH less than 5.6. Normal rainwater is saturated with CO₂ and has a pH of 5.6. The pH of acid rain can be as low as 2.1, lower than vinegar or lemon juice. pH 7 is neutral, not acidic.

Q7. Ground-level ozone is classified as a: ✓ **B — secondary pollutant formed by reactions of NO_x and VOCs in sunlight** **1 mark**

Ground-level ozone is a secondary pollutant; it is not emitted directly but forms when sunlight drives photochemical reactions between nitrogen oxides and volatile organic compounds (VOCs). It should not be confused with stratospheric ozone (option C describes stratospheric ozone's beneficial role). It is also not a significant greenhouse gas — that is option D's misleading framing.

Q8. The catalysts used in catalytic converters include: ✓ **B — platinum, palladium, and rhodium**

1 mark

Catalytic converters use metals **platinum (Pt), palladium (Pd), and rhodium (Rh)** as catalysts. These are precious metals that facilitate conversion of CO and NO_x into N₂, CO₂, and H₂O. Iron and nickel are common industrial catalysts but not used in catalytic converters.

Q9. Which gas is primarily responsible for both acid rain AND global warming? ✓ **B — SO₂** **1 mark**

SO₂ causes acid rain (by forming H₂SO₄) and is listed in Table 11.1 as also contributing to the greenhouse effect. CO₂ (option C) causes global warming but is not primarily responsible for acid rain. Ozone is a secondary pollutant. N₂ is inert and causes neither.

Q10. Methane enters the atmosphere from livestock mainly through: ✓ **C — belching and flatulence during digestion** **1 mark**

The textbook states that animals such as cows, goats, and sheep have a unique digestive system that produces large amounts of methane, released through **belching and flatulence**. Wetlands, rice fields, and landfills are other methane sources, but combustion of fossil fuels (option D) does not specifically produce methane from livestock.

Q11. Which reaction correctly shows SO₃ forming sulphuric acid? ✓ **B — SO₃ + H₂O → H₂SO₄** **1 mark**

The textbook equation is: **SO_{3(g)} + H₂O_(l) → H₂SO_{4(aq)}**. Option A uses H₂ instead of H₂O (incorrect). Option C uses O₂ (incorrect reactant). Option D produces sulphurous acid H₂SO₃, which is formed from SO₂ + H₂O, not SO₃.

Q12. Photosynthesis removes which pollutant from the atmosphere? ✓ **C — CO₂** **1 mark**

Photosynthesis: 6CO₂ + 6H₂O → C₆H₁₂O₆ + 6O₂. Plants use **carbon dioxide** and water to produce glucose and oxygen. SO₂, NO₂, and CO are not substrates for photosynthesis; they are removed by other environmental or engineered processes.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) Composition of clean, dry air **3 marks**

Air is a **mixture of gases**. The composition of clean, dry air by volume is: **1**

Gas	Approximate % by Volume
Nitrogen (N ₂)	78%
Oxygen (O ₂)	21%
Argon (Ar)	0.93%
Carbon dioxide (CO ₂)	0.038%
Others (Ne, He, CH ₄ , Kr, H ₂)	trace

The two most abundant gases are **nitrogen (78%)** and **oxygen (21%)**. **1** The remainder consists of noble gases and carbon dioxide. **1**

OR

2(i) OR — Atmosphere, water vapour, and variation **3 marks**

The **atmosphere** is the envelope of gases and water vapour surrounding planet Earth. **1** **Water vapour** is gaseous water present in the air; it is not a fixed component. **1** The amount of water vapour varies because it depends on temperature and the availability of water bodies: over deserts it is very low, while in tropical rainforests it can reach up to 4% of air by volume. Temperature, altitude, and proximity to oceans all affect water vapour content. **1**

2(ii) Pollutants and their effects 3 marks

A **pollutant** is anything in air, water, or soil that has a harmful effect on some part of the environment. 1

Four major air pollutants and their effects: 1

- **SO₂**: causes breathing difficulties, bronchitis, acid rain.
- **NO_x**: causes photochemical smog, acid rain, respiratory problems.
- **CO**: combines with haemoglobin; causes brain damage and death.
- **CO₂**: increases global warming, contributing to climate change.

1 mark for naming any 4 with correct effects

OR

2(ii) OR — Natural vs human sources 3 marks

Natural sources are processes not caused by humans: 1 (i) forest fires release smoke and CO₂; (ii) volcanic eruptions emit SO₂, ash, and poisonous gases; (iii) electrical discharges (lightning) produce nitrogen oxides. 1 **Human activity sources**: (i) burning fossil fuels in vehicles and power plants releases SO₂, NO_x, CO; (ii) deforestation reduces CO₂ absorption; (iii) incineration of waste produces smoke; (iv) industrial units release various pollutants. 1

2(iii) Properties and effects of SO₂ 3 marks

Physical properties of SO₂: colourless gas with an unpleasant and irritating odour; soluble in water. 1

Sources: burning of fossil fuels in power stations and industries. 1 **Harmful effects on human health**: readily absorbed in the respiratory system; acts as a powerful irritant; aggravates symptoms of people suffering from asthma, bronchitis, and emphysema; also responsible for acid rain and haze. 1

OR

2(iii) OR — Balanced equations for combustion reactions 3 marks

(a) $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO(g)}$ [1 mark] (b) $\text{S(s)} + \text{O}_2\text{(g)} \rightarrow \text{SO}_2\text{(g)}$ [1 mark] (c) $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$ [1 mark]

These reactions occur during the burning of fossil fuels; (a) shows incomplete combustion producing toxic CO; (b) shows sulphur combustion producing SO₂; (c) shows nitrogen fixation in car engines at high temperature.

2(iv) Carbon monoxide: properties and toxicity 3 marks

Carbon monoxide (CO) is produced by incomplete combustion when there is not enough oxygen. 1 It is **colourless and odourless**, so it cannot be detected by the senses, making it extremely dangerous. 1

When inhaled, CO combines with haemoglobin to form **carboxyhaemoglobin**, which cannot carry oxygen. This prevents blood cells from absorbing oxygen, causing unconsciousness, suffocation, and ultimately death. 1

OR

2(iv) OR — Particulate matter 3 marks

Particulates (particulate matter) are fine particles suspended in the air, including soot, smoke, dust, and other solid or liquid substances. 1 Three examples: (i) soot from incomplete combustion; (ii) smoke from industrial units; (iii) dust from road surfaces and construction. 1 **Harmful effects**: particles that penetrate deep into the lungs cause respiratory problems and lung cancer; they also increase the risk of cardiovascular disease. 1

2(v) Greenhouse effect and greenhouse gases 3 marks

The **greenhouse effect** is the process by which greenhouse gases in the lower atmosphere trap heat energy, reducing thermal energy loss to space and helping to regulate the Earth's temperature. 1 Three greenhouse gases: **water vapour (H₂O)**, **methane (CH₄)**, and **carbon dioxide (CO₂)**. 1 These gases absorb infrared radiation emitted by the Earth's surface and re-radiate some of it back downward, acting like a blanket around the planet. Without this effect, the Earth would be much colder. 1

OR

2(v) OR — Global warming and its consequences 3 marks

Global warming is the warming of the atmosphere due to human influence on the greenhouse effect, caused by disturbing the natural balance of greenhouse gas concentrations. 1 Four consequences if global warming continues: 2 — award 0.5 per valid point

- Temperature of the Earth will gradually increase.
- Climate change will affect rates of rainfall, causing flooding in some regions and drought in others.
- Polar ice will melt, causing a significant rise in sea levels.
- The atmosphere becomes hotter overall.

2(vi) Methane: sources and warming role 3 marks

Methane (CH₄) is a greenhouse gas that enters the atmosphere from a variety of sources. 1 Three sources: (i) decomposition of vegetation and waste in wetlands and landfills; (ii) rice fields; (iii) livestock (cows, goats, sheep) produce methane during digestion, released through belching and flatulence. 1 Higher levels of methane in the atmosphere **contribute to global warming** because methane absorbs and re-radiates infrared radiation, enhancing the greenhouse effect, which causes climate change. 1

OR

2(vi) OR — Ground-level ozone 3 marks

Ground-level ozone is often called smog. It is a secondary pollutant. 1 It is formed by complex photochemical reactions between **nitrogen oxides (NO_x)** and **volatile organic compounds (VOCs)** under the influence of sunlight. Sunlight breaks NO₂ into NO and atomic oxygen; the atomic oxygen then reacts with O₂ to form O₃. 1 Harmful effects: (i) irritates the respiratory organs, causes asthma and respiratory problems; (ii) damages vegetation, reduces crop yield, and damages forests. 1

2(vii) Acid rain: definition, pH, and responsible acids 3 marks

Acid rain is defined as rain having a pH less than 5.6. 1 Normal rainwater is saturated with CO₂ and has a pH of 5.6. Acid rain can have a pH as low as 2.1. 1 The two main acids responsible are **sulphuric acid (H₂SO₄)**, formed from SO₂ in the atmosphere, and **nitric acid (HNO₃)**, formed from NO₂. 1

OR

2(vii) OR — Conversion of SO₂ to sulphuric acid 3 marks

Step 1: Sulphur burns to form SO₂ $S(s) + O_2(g) \rightarrow SO_2(g)$ [0.5 mark] Step 2: SO₂ is oxidised to SO₃ $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ [1 mark] Step 3: SO₃ dissolves in water to form H₂SO₄ $SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$ [1 mark] Also: $4NO_2(g) + O_2(g) + 2H_2O(l) \rightarrow 4HNO_3(aq)$ [0.5 mark]

Nitrogen dioxide plays a catalytic role in the oxidation of SO₂ to SO₃ in the atmosphere, accelerating acid formation.

2(viii) Effects of acid rain on marble and aquatic life 3 marks

(a) **Marble/stone structures:** acid rain reacts with calcium carbonate (CaCO_3) in marble, disintegrating buildings and statues. 1

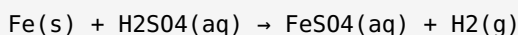


1 mark for equation (b) **Aquatic life:** acid rain makes lakes and rivers too acidic for fish and other aquatic organisms to survive, killing fish and destroying ecosystems. 1

OR

2(viii) OR — Effect on metals and comparison with normal rain 3 marks

Acid rain corrodes metals because sulphuric acid reacts with iron: 1



1 mark for equation Acid rain is more dangerous than normal rain because its pH can be as low as 2.1 (compared to 5.6 for normal rain), making it highly corrosive. It destroys trees by leaching minerals from soil, kills aquatic life, damages stone structures, and corrodes metals — effects that normal rainwater at pH 5.6 does not cause. 1

2(ix) Catalytic converters 3 marks

A **catalytic converter** is a device installed in the exhaust system of cars that converts harmful gases into less harmful substances. 1 The catalysts used are **platinum (Pt), palladium (Pd), and rhodium (Rh)**.

1 The main conversion equation is: 1



The converter also transforms CO to CO_2 , NO to N_2 and O_2 , and unburned hydrocarbons to CO_2 and H_2O .

OR

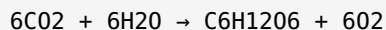
2(ix) OR — Strategies to reduce acid rain 3 marks

Three strategies: 1 per strategy

1. **Emission control:** reducing SO_2 and NO_x emissions from power plants, industries, and vehicles through anti-pollution technologies and cleaner fuels.
2. **Flue gas desulphurization:** wet or dry scrubbers in power plants remove SO_2 from flue gases before they are released into the atmosphere. Calcium oxide (CaO) is used to absorb SO_2 .
3. **Low sulphur fuels:** promoting fuels with low sulphur content minimises the amount of sulphuric acid formed in the atmosphere.

2(x) Photosynthesis and its role in reducing CO₂ 3 marks

Photosynthesis is a biochemical process in which plants use solar energy to convert carbon dioxide and water into glucose in the presence of chlorophyll. 1



1 mark for equation Photosynthesis removes CO₂ from the atmosphere and turns it into organic matter, thereby maintaining the carbon cycle and helping to counteract the build-up of this greenhouse gas that drives global warming. Forests act as carbon sinks. 1

OR

2(x) OR — Strategies to reduce global warming (climate change) 3 marks

Three strategies (1 mark each):

1. **Transition to renewable energy:** increasing wind, solar, hydro, and geothermal energy reduces reliance on fossil fuels and cuts greenhouse gas emissions.
2. **Afforestation and reforestation:** planting trees increases CO₂ absorption through photosynthesis, reduces atmospheric CO₂, and improves biodiversity.
3. **Reduce CO₂ from transport:** promoting electric vehicles and public transport reduces burning of fossil fuels, cutting CO₂ and NO_x emissions.

2(xi) Personal protective measures against air pollution 3 marks

(a) **N95 masks** are designed to filter out airborne particles including dust, smoke, and some airborne chemicals. They block at least 95% of particles, protecting the respiratory system from particulate matter and some chemical pollutants. 1.5 (b) **Carbon monoxide detectors** are installed in homes and offices to alert people to dangerous CO levels. Since CO is colourless and odourless, it cannot be detected by senses. The detector allows quick action: ventilating the area, sealing off sources, or evacuating. 1.5

OR

2(xi) OR — Air quality index 3 marks

An **air quality index (AQI)** provides information about the general air quality of a location. 1 It uses a numerical scale or colour-coded system to indicate the level of air pollution, from clean (green) to hazardous (red/purple). 1 Practical uses: (i) a person with asthma can check the AQI before going outdoors and choose to stay indoors on high-pollution days; (ii) schools can cancel outdoor activities when the AQI reaches unhealthy levels, reducing children's exposure to ozone and particulates. 1

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) Six major air pollutants and primary vs secondary distinction 5 marks

(a) The six major air pollutants: 4 marks — 4 points awarded for any 6 correct entries

Pollutant	Main Source	One Harmful Effect
SO_x (SO ₂ , SO ₃)	Burning fossil fuels (power plants, vehicles)	Breathing difficulties; acid rain; haze
NO_x (NO, NO ₂)	Car engines, power plants at high temperature	Acid rain; photochemical smog; lung disease
CO	Incomplete combustion of fuels and wood	Forms carboxyhaemoglobin; causes death
CO₂	Burning fossil fuels; deforestation	Enhances greenhouse effect; global warming
CH₄	Livestock digestion; wetlands; landfills	Greenhouse gas; contributes to global warming
Ground-level ozone	Photochemical reaction of NO _x + VOCs	Respiratory irritation; damages crops/forests

(b) A **primary pollutant** is emitted directly from a source (e.g., CO from vehicle exhaust). A **secondary pollutant** is not emitted directly but forms in the atmosphere through chemical reactions (e.g., ground-level ozone forms from NO_x + VOCs in sunlight). 1

OR**3(i) OR — Sources and combustion equations** 5 marks

(a) **Natural sources** of air pollution: (i) forest fires release smoke and CO₂; (ii) volcanoes emit SO₂, dust, and poisonous gases; (iii) electrical discharges (lightning) produce NO_x; termites and cows also release methane naturally. 1.5 **Human activity sources:** (i) burning fossil fuels (coal, petroleum, natural gas) in power plants; (ii) vehicle engines burning petrol/diesel; (iii) factories and industrial units; (iv) incineration of solid waste releases smoke, SO_x, and NO_x. 1.5 (b) Balanced equations: 2

(i) $\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO(g)}$ [limited] $\rightarrow \text{CO(g)}$ [incomplete combustion] (ii) $\text{S(s)} + \text{O}_2\text{(g)} \rightarrow \text{SO}_2\text{(g)}$ (iii) $\text{N}_2\text{(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO(g)}$ (iv) $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$

3(ii) Greenhouse effect and global warming in detail 5 marks

(a) **Greenhouse effect explained:** 3 When sunlight reaches the Earth, some is absorbed by the surface (vegetation, oceans, land) and warms it. As the surface heats up, it emits energy as **infrared radiation**. **Greenhouse gases** (water vapour, CO₂, CH₄) absorb this infrared energy. They then re-radiate some of it back towards the Earth's surface. This traps heat in the lower atmosphere, like a blanket. The process is a **natural phenomenon** of the Earth's energy distribution mechanism. Without it, the Earth would be too cold to support life (Fig. 11.3 in textbook). The higher the concentration of greenhouse gases, the greater the greenhouse effect and the greater the temperature increase. (b) **Global warming** is the warming of the atmosphere due to human influence on the greenhouse effect — specifically, disturbing the natural balance of greenhouse gas concentrations. 1 Human activities that intensify it: burning fossil fuels (increases CO₂), deforestation (reduces CO₂ absorption), industrial processes, and livestock farming (increases CH₄). Three consequences: 1 gradual temperature rise; polar ice melting causing sea level rise; changing rainfall patterns causing flooding and drought.

OR

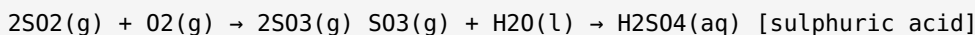
3(ii) OR — Strategies to reduce climate change and Paris Agreement 5 marks

(a) Three strategies: 3

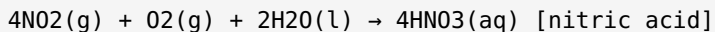
1. **Transition to renewable energy:** increasing use of wind, solar, hydro, and geothermal energy sources reduces greenhouse gas emissions from fossil fuel burning.
 2. **Afforestation and reforestation:** planting trees increases CO₂ absorption through photosynthesis, acts as a carbon sink, prevents soil erosion, and increases biodiversity. Forest regeneration rehabilitates damaged forests.
 3. **Reducing CO₂ from transport:** promoting electric vehicles and public transport systems reduces burning of fossil fuels and cuts CO₂ and NO_x emissions significantly.
- (b) The **Paris Agreement** is an international treaty to combat climate change through global cooperation. Its two key temperature targets are: (i) limit global warming to **well below 2°C** above pre-industrial levels; (ii) pursue efforts to limit it to **1.5°C**. 2

3(iii) Acid rain: formation, chemistry, and effects 5 marks

(a) **Acid rain** is defined as rain having a pH less than 5.6. Normal rainwater has pH 5.6 due to dissolved CO_2 . Acid rain forms when SO_2 and NO_2 from burning fossil fuels dissolve in rainwater. **1 Full sequence for SO_2 converting to H_2SO_4 : 1**



Conversion of NO_2 to HNO_3 : 1



NO_2 also acts catalytically in the oxidation of SO_2 to SO_3 . (b) Four harmful effects:

2 — 0.5 per effect with equation

- **Marble/stone:** $\text{CaCO}_{3(\text{s})} + \text{H}_2\text{SO}_{4(\text{aq})} \rightarrow \text{CaSO}_{4(\text{aq})} + \text{H}_2\text{O} + \text{CO}_2$ — statues and buildings disintegrate.
- **Iron metals:** $\text{Fe}_{(\text{s})} + \text{H}_2\text{SO}_{4(\text{aq})} \rightarrow \text{FeSO}_{4(\text{aq})} + \text{H}_{2(\text{g})}$ — metals corrode rapidly.
- **Aquatic life:** lakes become too acidic for fish to survive; fish are killed and ecosystems destroyed.
- **Trees:** acid rain leaches nutrients from soil and directly damages leaves and bark, destroying forests.

OR

3(iii) OR — Strategies to solve acid rain and catalytic converters 5 marks

(a) Three strategies: **3**

1. **Emission control:** reducing SO_2 and NO_x from industrial units, power plants, and vehicles using anti-pollution technologies and cleaner fuels. This directly reduces the precursors of acid rain.
2. **Flue gas desulphurization:** wet or dry scrubbers in power plants remove SO_2 from flue gases before atmospheric release. Calcium oxide (CaO) absorbs SO_2 : $\text{CaO} + \text{SO}_2 \rightarrow \text{CaSO}_3$.
3. **Low sulphur fuels:** promoting use of fuels with low sulphur content minimises SO_2 formation and consequently reduces sulphuric acid formation in the atmosphere.

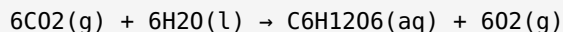
(b) **Catalytic converters** are installed in car exhaust systems. They convert NO and CO (acid rain and global warming precursors) into N_2 , CO_2 , and H_2O : **2**



Catalysts: **platinum, palladium, and rhodium**. They are effective because they lower the activation energy for these reactions without being consumed. This prevents NO from forming HNO_3 (a component of acid rain).

3(iv) Photosynthesis, carbon cycle, and risk factors 5 marks

(a) **Photosynthesis** is the biochemical process in which plants use solar energy to convert CO₂ and water into glucose in the presence of chlorophyll, producing oxygen. 1



1 mark for equation Photosynthesis plays a key role in the **carbon cycle** by removing CO₂ from the atmosphere and converting it into organic matter. Forests act as major carbon sinks. Deforestation therefore increases atmospheric CO₂, while afforestation reduces it. This makes photosynthesis an essential natural counterbalance to fossil fuel combustion. 1 (b) Three risk factors and linked diseases: 2

- **Long-term exposure to pollutants:** breathing air containing NO_x, SO₂, particulates, and ozone damages the respiratory system and contributes to asthma, lung cancer, and chronic obstructive pulmonary disease (COPD).
- **Smoking:** tobacco smoke damages the lungs and airways, increasing the risk of COPD and lung cancer; cigarette smoke also harms non-smokers nearby.
- **Indoor pollution:** poor ventilation, mould, dust, mites, and household chemicals cause respiratory problems, allergies, and asthma.

OR

3(iv) OR — Catalytic converters and personal protective measures 5 marks

(a) **Catalytic converters** are installed in the exhaust system of cars to reduce harmful emissions. 1 Three catalysts used: **platinum (Pt), palladium (Pd), and rhodium (Rh)**. 1 Main conversion:



Harmful gases converted: (i) CO → CO₂; (ii) NO → N₂ and O₂; (iii) unburned hydrocarbons → CO₂ and H₂O.

These reactions reduce both acid rain precursors and toxins from car exhaust. 1 (b) Three personal protective measures: 2

1. **N95 masks:** filter out at least 95% of airborne particles including soot, smoke, dust, and some chemicals; protect the respiratory system from particulate matter during outdoor pollution events.
2. **Gas masks with activated carbon filters:** absorb certain gases and volatile organic compounds; used in high-pollution environments or industrial settings.
3. **Carbon monoxide detectors:** installed in homes and offices; alert occupants to dangerous CO levels (which cannot be smelled), allowing quick ventilation or evacuation to prevent poisoning.